

RAHUL MITRA

OPERATIONS RESEARCH ENGINEER PROFILE

Singapore | mitrarahul2002@gmail.com | [linkedin.com/in/rahulmitra-dev](https://www.linkedin.com/in/rahulmitra-dev) | github.com/Rah-Rah-Mitra | rahul-mitra.com

PROFILE SUMMARY

Early-career operations research engineer combining NUS ISE foundations with deployable software, simulation, optimization, and analytics automation for scheduling, digital twin, process improvement, and decision-support roles.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Operations Research; Simulation; Statistics for Engineering Applications; Industrial Systems Engineering; Artificial Intelligence; 3D Computer Vision; Software Engineering
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

Data Analytics and Project/Product Management Intern, Abbott Laboratories

Jan 2026 - Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling; compared heuristic, MIP, and genetic-algorithm approaches and researched robust optimization.
- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of previously unseen raw data with zero execution failures.
- Productionized and supported a team-built APC simulator for live manufacturing and engineer training by packaging it in Docker and deploying it to Azure App Service.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

SELECTED PROJECTS AND COMPETITIONS

Hybrid Flow Shop Digital Twin Optimizer

2026

- Modeled hybrid flow-shop scheduling as a SimPy digital twin with explicit objectives and constraints, using CP-SAT, simulation, and deployment-oriented Python tooling.

Volt Pulse SG - SMU Hack For Cities Finalist

Jan 2026

- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

IE2110 Operations Research Graph Optimization

2024

- Applied IE2110 Operations Research I concepts from linear programming, network flow models, and nonlinear programming; implemented Jupyter SSSP/APSP graph optimization for engineering decision support.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: CP-SAT; Hybrid Flow-Shop Scheduling; SimPy; Robust Optimization Research; Digital Twin Simulation; Objective Functions & Constraints; Graph Optimization; Network Flow; Dijkstra; Python; Statistics; Excel VBA; Decision Analytics.

- Excel 2019 Advanced - analytics automation and spreadsheet modeling.
- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.